

2006ICT Object Oriented Software Development

Assignment Specification: Milestone 1

Assignment Title: Tetris Game Development (or Approved Alternative System)

Submission: Milestone 1

Due Date: End of Friday of Week 5 (21/08/2026)

Objective

The primary objective of this assignment is to utilize Object-Oriented Software Development principles and the Java programming language to develop a system that demonstrates good design and implementation practices. By default, students are expected to develop an enhanced version of the classic Tetris game.

However, if a group prefers to develop an alternative system, the following conditions apply:

- The group must submit a **proposal** that clearly describes the alternative system.
- The proposal must demonstrate that the system's complexity and difficulty level are comparable to the Tetris game.
- The proposal must identify and map specific features of the alternative system to the expected functions (features) of the Tetris game.
- The proposal must be submitted to both the **lab tutor** and the **course convenor** by the **end of Week 2**.
- The proposal requires formal approval from the lab tutor and course convenor before the group can proceed with the alternative system.

Background

Tetris is a classic puzzle video game originally designed and programmed by Alexey Pajitnov in 1984. The game involves players manipulating tetrominoes, shapes composed of four square blocks each, which descend onto a playing field. The goal is to fit these shapes into complete rows, which then disappear, earning points for the player. The game ends when the tetrominoes stack up to the top of the playing field and no more shapes can enter.

General Functions

To understand the general requirements, please study the provided demo video for milestone one. This will give you a comprehensive understanding of the gameplay mechanics, user interface, and features required for your implementation.

Submission

The submission for this milestone is a Word or PDF document based on the template provided on the course website. The document must include the following:

1. GitHub Project Link:

- Provide the link to your GitHub project repository.
- Ensure that you have invited your **lab teacher** to your project so they can access it. You can ask your lab teacher for their GitHub ID in your lab session.

2. Demo Video Link:

- Include a link to a video that demonstrates your game or approved alternative system.
- Detailed instructions for the video will be provided later in this specification.

Note: Each group only needs to make one submission. A student (preferably the team leader) should submit the document on behalf of the entire group.

Note: Peer review is submitted separately from this document — see the **Peer Review** item under Penalties, and the standalone Peer Review Guidelines and Form on the course website.

Overall Marking Criteria

The assignment will be assessed based on the components and points outlined below. This provides a clear overview of how marks are distributed across different sections of the project:

#	Section	Title	Points
1	Project Management	Project Planning	5
2	Requirement Analysis	Functional Requirements (FR)	10
3	Requirement Analysis	Non-Functional Requirements (NFR)	5
4	Requirement Analysis	Use Case Diagram	5
5	Requirement Analysis	Activity Diagram	5
6	Implementation	GitHub	5
7	Implementation	GitHub Advanced	5
8	Implementation	File Structure	5
9	Implementation	Coding	10
10	Demonstration	Video Quality	5
11	Demonstration	Splash Window	5
12	Demonstration	Main Screen	5
13	Demonstration	Configuration Screen	5
14	Demonstration	High Score Screen	5
15	Demonstration	Game Play	5
16	Demonstration	Erase Full Rows	5
17	Demonstration	Pause Function	5
18	Demonstration	Exit Function	5

Total: 100 points

Detailed Marking Criteria

The following sections provide detailed descriptions and requirements for each of the above components.

Project Management

1. Project Planning (5 points)

- Provide a clear and organized project plan using the template provided on the course website.
- The plan should define the major tasks, assign responsibilities to group members, estimate time for each task, and track actual time spent.

- Include a summary of group meetings and the tools used for communication and project management.
- List all team members in your group, their roles, and their main duties. Roles are self-defined by your team based on how you organised your work (e.g., Project Manager, Programmer, Tester, Architect, Documentation Lead).

Requirement Analysis

2. Functional Requirements (FR) (10 points)

- Provide a list of functional requirements that correspond **only to the features demonstrated in the Milestone 1 demo video**.
- Each functional requirement should include a unique ID, a clear description, and follow the precondition-event-postcondition pattern.
- Requirements must be complete, consistent, unambiguous, and testable.

3. Non-Functional Requirements (NFR) (5 points)

- Provide at least one non-functional requirement for each of the **URPS categories** (Usability, Reliability, Performance, Supportability) in the FURPS+ model.
- Each NFR must have a unique ID, be testable, and clearly indicate its category.

4. Use Case Diagram (5 points)

- Create a use case diagram based on your functional requirements.
- The diagram must use correct UML notation and describe services provided to the user without implementation details.

5. Activity Diagram (5 points)

- Draw an activity diagram showing the flow from **starting a new game to ending a game**.
- The diagram must follow UML notation and include key activities (e.g., play game, pause, game over).
- There is no need to include configuration or high score updates in this diagram.

Implementation

6. GitHub (5 points)

- Provide the GitHub repository link along with a clear screenshot of the commit history.
- It is not necessary to include all commits, but include enough to demonstrate good version control practices. The total length of the screenshot should not exceed one page.
- Marks may be deducted for issues such as an insufficient number of commits, unclear or poor commit messages, or commits that are heavily clustered within a short time frame.

7. GitHub Advanced (5 points)

- Demonstrate the use of GitHub features, including pull requests, code reviews, and tags.
- Provide supporting screenshots as evidence. It is not necessary to include screenshots for every pull request or code review discussion, but include a representative selection to demonstrate that your group actively engaged in these activities.
- The total length of screenshots in this section should not exceed two pages.
- All group members must contribute to the use of GitHub and collaborative development.

8. File Structure (5 points)

- Provide a screenshot of your GitHub repository file structure.
- The structure must follow the standard **IntelliJ project layout with Maven build**.

- Ensure no nested projects (i.e., no project directory inside another project).

9. Coding (10 points)

- Provide code snippets (via screenshots) demonstrating your use of:
 - JavaFX
 - Enhanced for loops
 - Enhanced switch statements or expressions
 - Interfaces
 - Abstract classes
 - Java Record types
- Snippets must come from your actual project code, not from external examples.
- For each code snippet, please provide a short paragraph explaining its purpose and functionality.

Demonstration

The marking criteria in the Demonstration section will be assessed entirely based on your submitted video. Please note that you only need to provide the video link in your submission document — there is no need to upload the video file itself. You may host your video on a public platform such as YouTube. If anything in this section is unclear, please refer to the Milestone 1 demo video provided on the course website.

10. Video Quality (5 points)

- Submit a video (max 7 minutes) demonstrating your system.
- The video must be clear, narrated, and show all required features.

11. Splash Window (5 points)

- Show a splash window at startup with your group identity, course code, and other relevant information.
- The splash window should be centered and displayed for a few seconds.

12. Main Screen (5 points)

- Demonstrate the main screen containing at least **Play**, **Configuration**, **High Scores**, and **Exit** buttons.

13. Configuration Screen (5 points)

- Show a functional configuration screen with settings for field size, level, music, sound effects, AI play, and extended mode.
- The configuration controls should be fully interactive—for example, checkboxes should toggle selection when clicked, and sliders should display the current selected value as they are adjusted.
- There is no need to connect the configuration settings to gameplay functionality at Milestone One.
- A **Back** button must return to the main screen.

14. High Score Screen (5 points)

- Display a high score screen showing the top 10 scores.
- Using dummy data is acceptable, and it is not required to connect the scores with actual gameplay results at Milestone One.
- Include a **Back** button to return to the main screen.

15. Game Play (5 points)

- Demonstrate the game field (10x20), smooth tetromino movement, and basic controls (arrows for move/rotate).
- **Smooth** means the tetromino's on-screen position visibly transitions between two adjacent rows during its normal (automatic, timer-driven) downward movement — e.g., moving from row 1 to row 2 shows intermediate positions (row 1.1, 1.2, ... 1.9) rather than the block instantly disappearing from row 1 and reappearing at row 2 with no animation in between. This requirement applies only to normal downward movement; it does not apply when the player presses the down key to speed up or hard-drop the piece, where intermediate steps may not be visible. Submissions where the normal downward movement snaps straight from one row to the next with no visible transition may lose up to **3 points**.

16. Erase Full Rows (5 points)

- Show correct detection and erasure of full rows, including multiple-row erasure and proper color maintenance.

17. Pause Function (5 points)

- Show the game pausing/resuming with the **P** key, and display an appropriate paused message.

18. Exit Function (5 points)

- Show the **Exit** button confirmation dialog with working **Yes** (exit program) and **No** (return to main screen) options.

Penalties

1. Unprofessional documentation

- Always use the official Milestone 1 submission template provided on the course website, and complete **all required cover page fields** (Campus, Group Number, Group Members with Student ID/Name, Team Leader, Lab Teacher, Project Name). If your group has members enrolled in different campuses, enter the campus in which your Team Leader is enrolled.
- Submissions that do not use the provided template, that leave required cover page fields incomplete, or that otherwise have poor or inconsistent layout, spelling or grammatical errors, overly small fonts, or low-resolution images, may incur a penalty of up to **10 points**.

2. Peer Review

- Every student must individually submit a peer review report for Milestone 1, using the standalone Peer Review Form (see the Peer Review Guidelines on the course website). Peer review is **not** part of the Overall Marking Criteria above and is assessed entirely through this penalty item.
- **Failure to submit:** a student who does not submit their own peer review report for Milestone 1 will have **5 points** deducted from their Milestone 1 mark.
- **Rating-based adjustment:** a student's individual Milestone 1 mark may also be adjusted based on the average peer rating they receive from teammates for this round, in accordance with the mark-adjustment table in the Peer Review Guidelines.

3. Late submission

- Late submissions will receive a penalty in accordance with the University's late submission policy.

4. Lab progress checks (recommended, not mandatory)

- From the second lab session (Week 3), groups are **encouraged** (but not required) to report their progress to the lab teacher during the lab session.
- The **team leader** or a designated group representative may report the group's progress and any issues to the lab teacher, and seek feedback or clarification on the assignment or lab tasks.

- It is encouraged that students first discuss issues within their group before escalating to the lab teacher or course convenor. If issues cannot be resolved by the lab teacher, they will be forwarded to the course convenor.
- This is a support opportunity, not an assessment item: **no penalty applies** if a group does not report progress during a lab session.

5. Access issues

- Please ensure that your lab teacher can access your GitHub repository and demonstration video. If either is not accessible, you will receive **zero** for the relevant marking items.